

AFRILANG Stdlib

std/c/mlpca

Français. Bibliothèque AFRILANG 'std/c/mlpca' (niveau complex, catégorie complex). Elle expose 7 fonction(s) : pcaSum, pcaMean, pcaMin, pcaMax, pcaVar, pcaStd, pcaNorm.

```
'import "std/c/mlpca"' · 'use mlpca'
```

```
- 'pcaSum(v list number) → number' - 'pcaMean(v list number) → number' - 'pcaMin(v list number) → number' - 'pcaMax(v list number) → number' - 'pcaVar(v list number) → number' - 'pcaStd(v list number) → number' - 'pcaNorm(v list number) → list number'
```

English. AFRILANG library 'std/c/mlpca' (tier complex, category complex). Exports 7 function(s): pcaSum, pcaMean, pcaMin, pcaMax, pcaVar, pcaStd, pcaNorm.

```
'import "std/c/mlpca"' · 'use mlpca'
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- 'pcaSum(v list number) → number' - 'pcaMean(v list number) → number' - 'pcaMin(v list number) → number' - 'pcaMax(v list number) → number' - 'pcaVar(v list number) → number' - 'pcaStd(v list number) → number' - 'pcaNorm(v list number) → list number'
```

Contents

Français

1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/mlpca' (niveau complexe, catégorie complexe). Elle expose 7 fonction(s) : pcaSum, pcaMean, pcaMin, pcaMax, pcaVar, pcaStd, pcaNorm.

'import "std/c/mlpca"' · 'use mlpca'

```
- `pcaSum(v list number) → number`
- `pcaMean(v list number) → number`
- `pcaMin(v list number) → number`
- `pcaMax(v list number) → number`
- `pcaVar(v list number) → number`
- `pcaStd(v list number) → number`
- `pcaNorm(v list number) → list number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/mlpca"
use mlpca
```

Niveau : complexe · Catégorie : Modules complexe (c/)
Domaines avancés – graphes, NLP, réseaux complexes.

3. Référence API

- - pcaSum(v list number) → number
 - pcaMean(v list number) → number
 - pcaMin(v list number) → number
 - pcaMax(v list number) → number
 - pcaVar(v list number) → number
 - pcaStd(v list number) → number
 - pcaNorm(v list number) → list

4. Exemples d'utilisation

```
import "std/c/mlpca"
use mlpca
```

```
create result = pcaSum(/* args */)
say result
```

```
create result = pcaMean(/* args */)
say result
```

```
create result = pcaMin(/* args */)
say result
```

```
create result = pcaMax(/* args */)
say result
```

```
create result = pcaVar(/* args */)
say result
```

```
create result = pcaStd(/* args */)
say result
```

5. Bonnes pratiques

- Préférez ``import "std/c/mlpca"`` en tête de fichier.
- Lancez ``afrilang check`` avant de livrer.
- Combinez avec les modules stdlib de la même catégorie.
- Voir `/stdlib/` pour le source live et les démos playground.
- Signalez les problèmes sur GitHub si une export manque.

6. Annexe — source

module mlpca

```
function pcaSum(v list number) returns number
end
```

```
function pcaMean(v list number) returns number
end
```

```
function pcaMin(v list number) returns number
end
```

```
function pcaMax(v list number) returns number
end
```

```
function pcaVar(v list number) returns number
end
```

```
function pcaStd(v list number) returns number
end
```

```
function pcaNorm(v list number) returns list number
end
end
```

English

1. Overview

AFRILANG library 'std/c/mlpca' (tier complex, category complex). Exports 7 function(s): pcaSum, pcaMean, pcaMin, pcaMax, pcaVar, pcaStd, pcaNorm.

```
'import "std/c/mlpca"' · 'use mlpca'
```

```
- `pcaSum(v list number) → number`
- `pcaMean(v list number) → number`
- `pcaMin(v list number) → number`
- `pcaMax(v list number) → number`
- `pcaVar(v list number) → number`
- `pcaStd(v list number) → number`
- `pcaNorm(v list number) → list number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/mlpca"
use mlpca
```

Tier: complex · Category: Complex modules (c/)
Advanced domains – graphs, NLP, complex networks.

3. API reference

- - pcaSum(v list number) → number
 - pcaMean(v list number) → number
 - pcaMin(v list number) → number
 - pcaMax(v list number) → number
 - pcaVar(v list number) → number
 - pcaStd(v list number) → number
 - pcaNorm(v list number) → list

4. Usage examples

```
import "std/c/mlpca"
use mlpca
```

```
create result = pcaSum(/* args */)
say result
```

```
create result = pcaMean(/* args */)
say result
```

```
create result = pcaMin(/* args */)
say result
```

```
create result = pcaMax(/* args */)
say result
```

```
create result = pcaVar(/* args */)
say result
```

```
create result = pcaStd(/* args */)
say result
```

5. Best practices

- Prefer ``import "std/c/mlpca"`` at the top of your file.
- Run ``afriLang check`` before shipping.
- Combine with related stdlib modules from the same category.
- See `/stdlib/` for the live source and playground demos.
- Report issues on GitHub if an export is missing or undocumented.

6. Appendix — source

module mlpca

```
function pcaSum(v list number) returns number
end
```

```
function pcaMean(v list number) returns number
end
```

```
function pcaMin(v list number) returns number
end
```

```
function pcaMax(v list number) returns number
end
```

```
function pcaVar(v list number) returns number
end
```

```
function pcaStd(v list number) returns number
end
```

```
function pcaNorm(v list number) returns list number
end
end
```

Référence rapide / Quick reference

- Import : `import "std/c/mlpca"`
- Vérification : `afriLang check`
- Documentation : <https://afriLang.dev/stdlib/std/c/mlpca/>

Source (extrait)

```
module mlpca

function pcaSum(v list number) returns number
end

function pcaMean(v list number) returns number
```

```
end

function pcaMin(v list number) returns number
end

function pcaMax(v list number) returns number
end

function pcaVar(v list number) returns number
end

function pcaStd(v list number) returns number
end

function pcaNorm(v list number) returns list number
end
end
```